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Flexibility in short-term electricity markets for renewable integration and uncertainty mitigation: A comprehensive review

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ABSTRACT

Globally, the share of renewable energy sources in the electricity mix is increasing. However, higher levels of renewable energy sources and especially those with an intermittent nature introduce uncertainty in delivering reliable and secure electricity. Hence, flexibility becomes more important to counteract imbalances between generation and consumption. This review paper contributes to the nuanced ways in which flexibility can be strategically employed to navigate and mitigate uncertainties amid evolving market dynamics and increasing share of renewable sources in the energy mix. It investigates the impact of renewable energy sources on short-term electricity markets, with a specific focus on day-ahead, intraday, and balancing markets in the Central Western Europe region. It explores the design characteristics and parameters of these markets, emphasizing how these markets deal with the uncertainty arising from the limited predictability of renewable resources. In this context, flexibility becomes a crucial element in reducing this uncertainty. The primary objective is to fully understand and analyze the intricate interplay between the sequential short-term markets and the imperative for flexibility. First, the current short-term markets are discussed. Next, the paper examines how flexibility in its three dimensions (i.e., time, space, and demand-response) can strategically function to not only address but also proactively alleviate uncertainties within these markets. Advancements in market coupling, forecasting accuracy, and increased liquidity have significantly enhanced the efficiency of these markets, particularly in accommodating the growing presence of renewable energy sources.

1. Introduction

European electricity markets have rapidly transformed in recent years due to the increasing use of renewable energy sources (RES). The gradual replacement of conventional power plants by RES affects the characteristics of these markets, influencing their organization, products and electricity pricing [1]. Unlike commodities like oil or natural gas, electricity possesses unique characteristics such as grid-bound distribution and a lack of significant storability in large quantities [2]. Real-time balancing of supply and demand is essential due to the instantaneous nature of electricity generation and consumption [3]. Tied to this real-time balancing problem are geographical capacity and grid-bound constraints due to transmission losses and infrastructure limitations, leading to different pricing zones across Europe. Historically, electricity production relied on centralized thermal and hydro power plants, with fossil fuel plants providing flexibility to address fluctuations in demand. Market pricing was adjusted on hourly or half-hourly intervals to align supply and demand dynamics [4]. However, the rise of RES has led to increased mismatches between nominated generation and actual deliveries triggering more instabilities and prompting market adaptations such as shorter delivery intervals [5].

Market participants in Europe manage their assets through self-scheduling, driven by trading activities starting years prior to delivery and extending until real-time. This process unfolds through four sequential markets, starting with (1) unstandardized forward and standardized future markets. Sellers (i.e. power producers) and buyers (i.e. final customers' suppliers or retailers) of electricity can hedge risks by exchanging future or forward contracts traded in these markets [6]. These initial markets typically persist until one day before delivery, at which point the power exchange conducts its centrally organized (2) day-ahead (DA) market. Following the clearance of the DA market, (3) intra-day (ID) market trading remains available until close to real-time. Following the closure of the ID market gates, the

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responsibility for maintaining system balance falls to the transmission system operator (TSO) in (4) the balancing (IB) markets. To ensure this equilibrium, the TSO secures and activates reserve capacity from balance service providers (BSPs) on the procurement side, while also settling imbalanced positions with balance responsible parties (BRPs) on the settlement side [3,7].

By 2030, the integration of RES is expected to increase substantially in Europe. The North Sea is being transformed into the European energy hub. The four North Sea countries — Belgium, the Netherlands, Germany, and Denmark — jointly aspire to achieve 120 GW of offshore wind turbines by 2030, constituting a more than fivefold increase from the current capacity. Looking ahead to 2050, the ambition is set at 300 GW [8]. Globally, the share of electricity production from fossil fuels is projected to decline from 60% to 40%, necessitating greater reliance on RES [9]. Higher levels of RES make the system more vulnerable to rapid changes in frequency, introducing uncertainty for delivering reliable and secure electricity supply. As the share of RES increases in the energy system, flexibility services are required to maintain grid stability and service quality [4]. Due to their limited controllability and predictability, there is a growing demand for flexibility—namely, the capability to make both upward and downward power adjustments to counteract temporary imbalances between generation and consumption [3]. Short-term markets serve as platforms where market participants can promptly address fluctuations.

This paper focuses on the design of short-term markets to enhance flexibility, facilitating the integration of RES. Because they are close to real-time, these markets play a vital role in enabling rapid responses to fluctuations in RES generation levels, allowing market participants to adjust consumption or production promptly. How can the design of short-term electricity markets be enhanced to facilitate the seamless integration of RES by maximizing flexibility options? This question serves as a focal point for addressing the intricacies of market dynamics and implementing strategies that foster a more harmonious incorporation of renewable energy within short-term electricity markets. This paper offers three main contributions: (1) enhancing understanding of the short-term market dynamics, (2) proposing improvements to short-term market design to increase flexibility and (3) the use of flexibility in the market to mitigate uncertainty, particularly in the context of integrating RES. Market design proposals are put forward, to enhance market liquidity, reduce price volatility, and empower end-users to actively participate in the energy market. The focus extends beyond the IB market to include both DA and ID markets. The paper emphasizes the significance of proactive market design adjustments to incentivize flexibility and ensure grid stability amidst uncertainties.

The paper is structured as follows. It begins with a discussion on flexibility and conducts a literature review. The sequential short-term markets and their design are discussed in Section 2. In Section 3, the focus shifts to examining the relationship between short-term markets and flexibility. This section not only discusses the existing relationship but also proposes various solutions for potential market modifications to enhance flexibility. Finally, Section 4 gives directions for further research and concludes the paper.

1.1. Flexibility

Flexibility can help market participants to adjust their consumption or production patterns in response to changes in renewable energy availability, enabling effective grid management and ensuring a dependable power supply. Flexibility can be approached through three overarching methods: (1) spatial adjustment of demand and supply via interconnectors and transmission links to achieve local equilibrium, here coupled markets come into play; (2) temporal adjustment of demand and supply through storage mechanisms; or (3) achieving balance between demand and supply through the implementation of demand-response systems, often facilitated by price signals [4,10,11]. Today both the need and the scope for much more granular and differentiated price signals are already increasing. The flexibility of demand is steadily rising, while the expenses associated with transmitting these diverse price signals are decreasing, thanks in part to advancements in smart metering technologies. Moreover, advancements in computing power indicate the potential for faster price resolution, enabling the exploitation of smaller time intervals and broader geographical coverage [12].

1.2. Literature review

This paper advances the current literature by offering a comprehensive and system-oriented review of recent developments in European electricity market design, with particular attention to the introduction of new short-term market products (e.g., 15-min day-ahead contracts, flexible block orders) and updated auction mechanisms (e.g., intraday auctions, balancing market reforms). In contrast to previous reviews that tend to either focus narrowly on regulatory developments or treat flexibility in broad conceptual terms, this study systematically analyzes how these innovations affect the practical operation and coordination of the DA, ID, and IB markets.

Specifically, it investigates whether these new designs have contributed to improved short-term price formation, more accurate forecasting integration, reduced dependence on last-resort balancing interventions, and better spatial and temporal alignment of system operations. By evaluating both the design intent and the operational implications of recent reforms, the paper provides a more applied and market-centric perspective than earlier works.

Furthermore, this study distinguishes itself by explicitly comparing market design changes across the short-term trading chain and highlighting their implications for liquidity, distributed flexibility participation, and cross-border harmonization. As summarized in Table 1, the scope and depth of this review go beyond cataloguing reforms—it offers a critical assessment of how these developments may shape the resilience, efficiency, and adaptability of future electricity markets.

The increasing proportion of electricity generated by RES, coupled with evolving consumption patterns among end-users and heightened investment in storage infrastructure, is already reshaping the power system. This presents significant challenges to wholesale electricity markets, originally designed for dispatchable power plants and rigid demand profiles. The growing mismatch between the current market framework and the emerging electricity landscape creates new market inefficiencies within an already imperfect market [13]. This misalignment hampers efficient market operations, adversely affecting both producers and consumers. To address these inefficiencies, the research is divided into the following key areas: (1) flexibility, (2) short-term electricity market design, and (3) market design reform.

Flexibility is critical for managing the increasing volatility and uncertainty associated with high shares of renewable energy sources. We categorize flexibility into spatial, temporal, and price-based mechanisms, corresponding to the system's need to balance demand and supply across regions (Space), over time (Time), and through economic signals (Price signals). These categories directly inform our scenario design and market design assessment. At the distribution level, Ref. [14] analyzes how Distribution System Operators (DSOs) are transitioning from asset-heavy investments toward procurement of third-party flexibility. The study identifies key technical and institutional barriers that prevent broader deployment. While focused on long-term grid planning, these insights are highly relevant for short-term flexibility activation—an area this study addresses by examining how DA, ID, and IB markets can better coordinate to support local flexibility. For electricity storage, Ref. [15] develops a co-optimization model that captures arbitrage value across short-term markets in Central Western Europe (CWE). The study finds that flexibility value depends on market integration and price signal quality across timeframes. Ref. [16] extends this discussion by analyzing storage and demand-side resources

within a joint reserve and network-constrained market, further underscoring the need for coordinated design across flexibility dimensions. Our work builds on these insights by evaluating how short-term market design can be adapted to enhance signal coherence and support broader participation of storage and responsive loads. Flexibility from heat pumps is addressed in Refs. [17,18], which explore commercial and residential applications, respectively. These studies show that while flexibility potential is significant, barriers such as comfort constraints and low profit margins limit participation. Ref. [19] reinforces these findings through a comprehensive review of residential thermal energy storage, highlighting regulatory and infrastructure hurdles to wider adoption. This study builds on these findings by assessing how market rules — such as product duration, gate closure timing, and bidding options — can better accommodate distributed flexibility. Ref. [20] quantifies the revenue potential for flexible resources in the DA and ID markets in Germany, identifying two key value drivers: "immediacy" and "ramping capability". These findings directly support our analysis of market design parameters, particularly around temporal granularity and pricing responsiveness. System-level flexibility is further discussed in Ref. [21], which outlines a phased approach for scaling flexibility as renewable penetration increases. The study emphasizes that flexibility must be supported across daily, weekly, and seasonal timescales. This aligns with our assessment of how short-term market design can complement long-term system planning. Finally, Ref. [22] reviews prequalification requirements for small-scale flexibility participation in five European balancing markets. It reveals regulatory fragmentation that may exclude aggregators and smaller actors. This study incorporates those findings by exploring how harmonized access rules across DA, ID, and IB markets can unlock broader system value from decentralized resources. These studies provide a comprehensive foundation for evaluating flexibility as a system resource. This paper builds on their insights to propose market-level interventions that can improve the visibility, accessibility, and valuation of flexible capacity across all short-term market layers.

Short-term electricity markets are now critical tools for managing operational uncertainty, activating flexibility, and improving the efficiency of system dispatch. This section evaluates literature that investigates how their design affects both market behavior and system performance. Ref. [3] provides a comprehensive overview of market design in the CWE region, detailing how pricing rules, gate closures, and trading intervals shape flexibility outcomes. Ref. [23] complements this by analyzing how day-ahead market design can be structured to co-optimize energy and reserve procurement, a crucial feature for high-RES systems. Our analysis draws on these parameters when evaluating design levers that could improve cross-market responsiveness and reduce inefficiencies in real-time balancing. Refs. [24,25] both examine how intraday price volatility — amplified by RES uncertainty — creates challenges and opportunities. While flexible producers may benefit from this volatility, renewable generators face increasing balancing costs. Ref. [26] further supports this point by showing how reserve scheduling must adapt under wind uncertainty, reinforcing the need for tighter DA-ID-IB integration. These dynamics validate our emphasis on improving intraday market structures and interfaces. Liquidity remains a major concern. Ref. [7] critiques the mismatch between system needs and traded volumes in Germany's ID market, suggesting that market formats must evolve to support higher RES integration. Recent crises have prompted broader policy reflection. Ref. [27] proposes a hybrid model that maintains short-term price formation for operational efficiency while layering in long-term risk instruments such as Contract for Differences (CfDs). Our study considers this dual-structure model when assessing how market signals can guide both dispatch and investment under uncertainty. Ref. [28] shows how intraday trading can act as a decentralized balancing mechanism, reducing real-time imbalances. Ref. [29] adds a cross-border dimension, finding that integrating ID markets across Europe improves consumer cost outcomes, particularly when coupled with hydro storage. These studies directly inform our consideration of intraday market harmonization and regional coupling. Ref. [30] analyzes the impact of new EU balancing market rules, including voluntary "second-chance" bids. It finds that these changes improve market efficiency but may also shift strategic behavior into capacity markets. This reinforces our focus on designing complementary balancing and intraday products that reduce unintended distortions. Ref. [31] adds further operational evidence from Belgium, demonstrating how enhanced reserve dimensioning contributes to improved market responsiveness—especially in systems with high RES penetration. Together, these sources emphasize that short-term markets are not static mechanisms but dynamic platforms requiring continuous design evolution. This study builds on this literature by identifying actionable design features that can improve real-time flexibility while supporting broader market stability and integration goals.

While short-term markets address operational variability, systemic challenges — such as investment risk, spatial congestion, and long-term adequacy — require broader market design reform. This section reviews literature that addresses these structural issues, many of which are under active policy debate in the EU. Ref. [4] sets out economic principles for modernizing market design under high-RES conditions, including internalizing flexibility, updating network tariffs, and rethinking support schemes. Ref. [32] echoes this by arguing that legacy market structures cannot fully accommodate today's decentralized, digital, and demand-responsive power systems. Ref. [13] categorizes existing reform proposals and finds that no single measure is sufficient—future-proof market design will require a combination of structural changes. Multiple studies address persistent inefficiencies. Refs. [33,34] highlight institutional and design barriers to RES integration in Europe, including outdated support schemes and a lack of coordination between national and EU-level mechanisms. These findings directly inform our evaluation of how market design must evolve to avoid locking in legacy inefficiencies. Ref. [5] provides empirical evidence that reducing product duration to 15 min leads to more accurate price signals and increased renewable dispatch. This supports our recommendation for more granular trading intervals to improve market alignment with RES variability. The spatial dimension is addressed in Refs. [35,36], which advocate for a shift to nodal pricing in Europe. These studies argue that locational pricing can help reduce congestion, improve investment signals, and activate demand-side flexibility—an increasingly urgent priority given rising electrification and grid constraints. Ref. [37] supports this through a quantitative evaluation of nodal pricing benefits in Texas, reinforcing the feasibility and system-level gains of spatial reform. Ref. [27], previously discussed, supports a hybrid market architecture in which short-term efficiency is preserved alongside long-term risk mitigation. Ref. [38] focuses on price formation, showing how suppressed volatility in real-world clearing procedures can lead to underinvestment in flexible capacity. This is consistent with our own finding that price signals must be restored or supplemented by well-designed flexibility products. Ref. [39] brings in a U.S. perspective, offering insights into the benefits of intertemporal coordination and locational marginal pricing already in use elsewhere. Finally, Refs. [40,41] evaluate the performance of EU market coupling and interconnector utilization, finding improvements in cross-border efficiency but also identifying persistent challenges. Ref. [42] complements the access and participation debate by exploring how demand response bidding formats can be better structured to support wider inclusion—particularly for smaller or less conventional actors. These studies demonstrate that electricity market reform must be multi-layered—addressing pricing, risk, spatial coordination, and institutional alignment. This paper draws on their insights to propose a reform framework that links short-term market improvements with long-term policy instruments, helping ensure flexibility, reliability, and economic efficiency in high-RES power systems.

Table 1
Overview of current literature on European electricity market design and their scope.

Reference	Time	Space	Price signals (DR)	DA	ID	IMB	Integration RES	Organization	Market-clearing	Commodity	Policy	Location
(Sæle et al. 2023), [14]	x	x	x				x			x		Norway
(Tina et al. 2022), [17]			x		x							Italy
(Marijanovic et al. 2022), [18]			x		x							Italy
(Mays, 2021), [38]	x			x		x			x			US
(Pape, 2018), [24]			x	x	x		x					EU
(Märkle-Huß et al. 2018), [5]	x			x		x	x	x				EU
(Ovaere et al. 2023), [41]		x		x			x	x	x			CWE
(Peng & Poudineh, 2019), [34]							x	x		x		EU
(Jin et al. 2020), [43]							x	x		x		EU
(Madlener & Ruhnau, 2021), [25]	x	x	x	x	x							EU
(Fabra, 2023), [27]			x	x	x		x			x		EU
(Eicke & Schittekatte, 2022), [35]		x	x				x	x		x		EU
(Conejo & Sioshansi, 2018), [32]											x	US
(Weber, 2010), [7]			x	x		x	x			x		EU
(Newbery et al. 2018), [4]	x	x	x			x	x	x	x	x		EU
(Neuhoff et al. 2023), [36]			x				x		x			EU
(Montoya et al. 2020), [40]	x			x						x		EU
(Brijs et al. 2017), [3]	x	x	x	x	x	x			x			CWE
(Silva-Rodriguez et al. 2022), [13]			x	x	x	x	x	x	x	x		CWE
(Zalzar et al. 2020), [29]							x	x		x		EU
(Poplavskaya et al. 2021), [30]	x			x		x						EU
(Goutte & Vassilopoulos, 2019), [20]	x	x	x	x	x	x						EU
(Papaefthymiou & Dragoon, 2016), [21]	x	x	x			x			x			EU
(Hogan, 2019), [44]							x	x		x		US
(Brijs et al. 2019), [15]	x		x	x	x	x						CWE
(Scharff & Amelin, 2016), [28]				x		x						EU

2. A discussion of short-term electricity markets

Over the last decades, power systems moved from a centralized organization to new frameworks aiming to enhance competition. Initially, state-owned and vertically integrated companies were in charge of the management of the whole power system, from generation to retail. Then, aiming for privatizing the electricity supply sector and attracting new investors, deregulation processes have occurred worldwide [45]. Along with the introduction of competition, there was a shift from cost-based price regulation to market-oriented price formation, accompanied by the establishment of market platforms for short- and long-term electricity trading. In competitive power markets, the wholesale price is set by the generation costs of the marginal technology, known as the short-run marginal costs of the most expensive plant required to meet demand, following the merit order principle [46].

Electricity markets are characterized by their trading timeline. Medium/long-term markets (i.e., futures and forward market) facilitate trading on longer horizons to manage risk in wholesale power prices [47]. Futures contracts are traded in organized markets (i.e., European Energy Exchange), while forward contracts entail agreements between sellers committing to produce a set energy quantity and buyers responsible for consumption. Forward and futures trading typically involve coarser products, which entail the delivery of power over longer periods categorized into baseload or peak contracts shown in Fig. 1. As the delivery time approaches, volume risk is managed through progressively finer hedging strategies. For instance, a retailer may purchase a baseload contract for its annual average demand a year or two in advance. Several months ahead, this contract may be re-profiled into monthly peak and baseload contracts. Within the week ahead, these contracts may undergo multiple re-profiles as demand forecasts adjust, resulting in separate daily peak and baseload contracts [47]. Finally, short-term markets come into play. Positions are re-profiled into hourly or quarter-hourly spot products to align with the expected demand curve, a practice known as “adding shape”, as shown in Fig. 2. Usually, a part of the capacity of thermal plants is contracted in medium/long

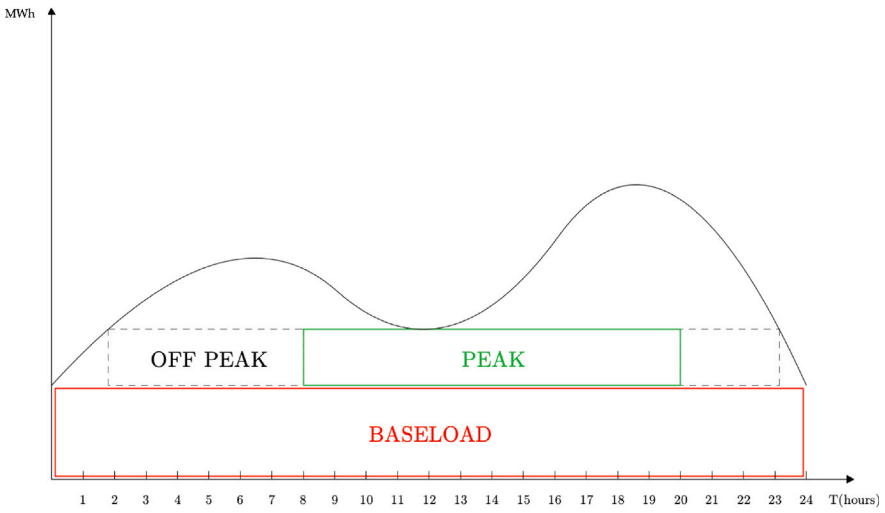


Fig. 1. Daily load profile (in MWh), coarser products in long term markets.

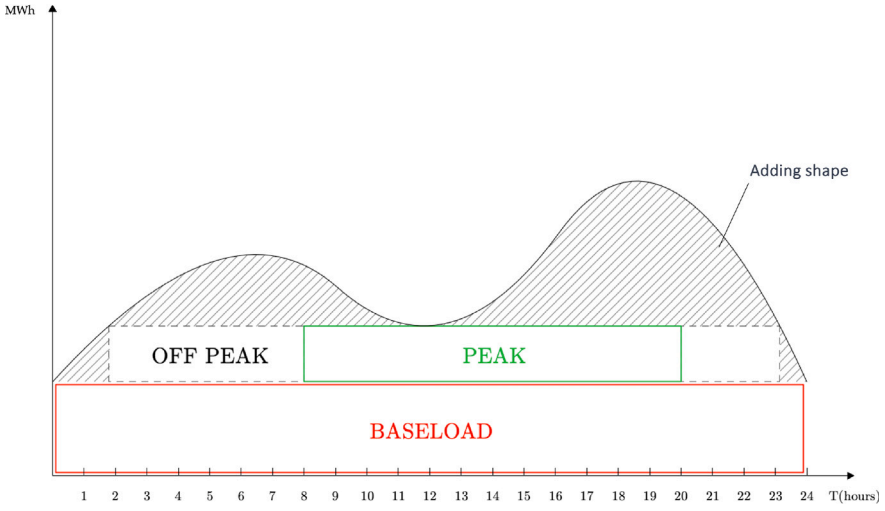


Fig. 2. Daily load profile (in MWh), adding shape in the short-term markets.

contracts, since these ensure fixed revenues for the producers, avoiding the uncertainties of short-term trading. RES exhibit stochastic behavior and are subject to limited predictability. Consequently, they are less suitable for long-term contracts, given the challenge of ensuring a specified production level well in advance of real-time operations [48].

A schematic overview of the sequence and timing of the various electricity markets is provided in Fig. 3. The short-term electricity markets play a important role in managing the dynamic nature of energy supply and demand. The following sections analyze the key components of these markets.

2.1. Day-ahead markets (DA)

DA markets conduct daily blind auctions for electricity delivery the following day. These markets, excluding Switzerland and Great Britain, are part of the Single Day-Ahead Coupling (SDAC), spanning 28 markets across Europe. The opening of the order book varies by bidding zone, ranging from D-45 (i.e., 45 days in advance of delivery) to D-2. Most markets close at 12:00 on D-1 (i.e., one day before delivery) [49]. Trading occurs hourly (semi-hourly in Great Britain) by submitting sell or purchase quantity/price bids before gate closure [50]. Following the provisions of EU Regulation 2019/943, all SDAC markets will launch 15-min Market Time Unit (MTU) products in all EU bidding zones starting from 11 June 2025 with delivery date of 12 June 2025 [51]. The price of a purchase bid indicates the highest price a buyer is willing to pay, while the price in a sell bid indicates the lowest price at which a seller is willing to sell. The intersection of the aggregated demand and supply curve determines the market-clearing price (MCP) and volume (MCV) at which the market is cleared [3]. This is done by a single price coupling algorithm, EUPHEMIA (Pan-European Hybrid Electricity Market Integration Algorithm). All buyers who submitted volumes at a price higher than the MCP have their volumes executed at MCP, while all sellers who offered volumes at a price lower than the MCP are also executed for those volumes at MCP [3,52,53].

Price Coupling of Regions (PCR) is a collaborative initiative among Nominated Electricity Market Operators (NEMOs) [49] aimed at establishing a unified solution for calculating electricity prices across Europe. It takes into account interconnection capacity constraints between pricing zones on a DA basis. All bids from participating exchanges are integrated into the same market-clearing algorithm to optimize available interconnection capacity. Price disparities between zones indicate congestion on interconnections, signaling full utilization of available capacity. An integrated European electricity market offers benefits such as enhanced liquidity, efficiency, and social welfare, leading to a more harmonized market. Market coupling distributes variability across a wider area, smoothing out non-correlated power fluctuations and leveraging interconnection capacity to mitigate local system variability,

facilitating the procurement of cost-effective flexibility from abroad [40,52].

The DA market features an hourly resolution, offering 24 hourly contracts during the auction, each corresponding to the 24 h of the following day. Contracts can be traded either in single hours (linear orders) or in blocks of combined hours to adjust positions from previously held forward and futures markets. A linear order can contain up to 256 price/quantity combinations for each delivery period of the following day. Furthermore, price-independent orders can be placed with identical quantities within a specified range, indicating the maximum and minimum price, to ensure execution at any price. Block orders, on the other hand, provide the opportunity to submit a volume profile for multiple delivery periods, which will be executed at a common price. Flexibility in the execution ratio can be applied to the volume profile. Smart and large blocks provide market participants with the opportunity to optimize their strategies, due to their complexity and size. These smart orders enable the implicit consideration of cost non-convexities and intertemporal dependencies. They account for various factors such as start-up and shutdown costs, ramp rates, minimum load levels, minimum up and down times, and energy buffer dynamics. An overview of all the supported SDAC measures can be found in [54]. Additionally, they offer a mechanism for more resilient price bidding, particularly in situations of price uncertainty [3,55]. They enable market participants to navigate complex dynamics, optimize strategies, and enhance flexibility in electricity markets.

In the DA auctions within the CWE region, prices are bounded by a floor of −500 €/MWh and a cap of 4000 €/MWh [50]. Price caps are typically enforced to prevent generators from setting excessively high prices, particularly when demand is price inelastic, which limit market power in the short run. However, they give rise to the issue of "missing money" where prices fail to fully capture scarcity during tight market conditions, diminishing profitability and resulting in underinvestment in capacity over the long run [56]. With the extensive integration of variable RES in the generation mix, conventional power plants are currently experiencing declining profitability due to fewer operating hours and lower electricity prices [57]. Therefore, the imposed price cap should be sufficiently high to enable these conventional power plants to recoup their fixed investment costs over a reduced number of operating hours. Negative prices arise when conventional generators are willing to pay to generate electricity to avoid expensive shutdowns or downward ramping, and when renewable generators are willing to pay up to the amount of subsidies they receive in order to generate electricity [58].

2.2. Intraday markets (ID)

After the DA market-clearing process, each BRP must provide a balanced position to the TSO, called "nominations", detailing planned schedules for individual units. This position includes DA market bids and volumes from futures and forward markets. The position taken in the DA market can undergo changes due to more accurate forecasts approaching real-time. These positions can be adjusted by conducting trades in the ID market. It is the final opportunity for market-based transactions. Despite being the smallest market in terms of trading volume, the ID market gains importance, particularly with the growing presence of RES in the energy mix. It serves various purposes, including fine-tuning positions closer to delivery or with more precision, managing forecast errors or unexpected occurrences, and facilitating advanced arbitrage and trading strategies. The ID market offers finer granularity and enhanced flexibility, allowing for further optimization of the portfolio. Without ID trading, the quantity differences between the DA position and the real-time production will be spilled into the IB market. A sufficiently liquid ID market can lower market participants' exposure to imbalance risks and financial penalties. Consequently, the availability of ID trading has the potential to transfer a portion of the flexibility requirement from real-time to the ID [24,57].

In the ID market, two distinct market types exist: continuous trading and auction trading, each with unique characteristics.

The continuous ID market trading in Europe starts at 15:00 D-1 (01:00 in GB and 14:00 in the Nordics). Lead times vary across bidding zones, ranging from 60 min in some countries to as little as 15 min in others prior to delivery [59]. Participants submit supply and demand bids to a central platform, where matching bids are continuously cleared on an individual basis. Three contract types — 60, 30, and 15 min — are available, allowing market participants to adjust positions from the hourly DA market to the 15 min IB market [60,61]. The order book, accessible to all participants, contains all submitted bids awaiting clearance. Participants can cancel non-cleared bids at any time. Bids in continuous trading are matched based on a price time priority: orders are matched according to the attractiveness of their price, with the time of submission to the central platform acting as a tiebreaker if there are two identical price offers. Consequently, a continuous trading ID market may yield varying prices for each trade, with the price determined by the bid that initiated the match, which may be referred to as pay-as-bid [57]. In the ID continuous trading, prices are constrained by a price floor of −9999 €/MWh and a price cap of 9999 €/MWh [60].

With increasing share of RES, ID auctions are key in reinforcing the ID reference price, optimizing the short-term markets. These auctions provide additional contract granularity and balancing opportunities. Introduced in Germany in 2014, ID auctions have seen significant growth in liquidity since then, with expansions to Great Britain, Switzerland, and other countries. By 2020, ID auctions were extended to include Austria, Belgium, France, and The Netherlands. Operating similarly to the DA market, participants submit both sell and purchase bids, with a MCP determined at the intersection of the aggregated supply and demand curve [62]. Each local auction has unique closing times and available contracts. Belgium, Austria, Germany, and the Netherlands offer 15 min contracts, closing daily at 15:00 D-1, while France provides 30 min contracts, closing daily at 17:00 D-1. Additionally, from June 13, 2024, NEMOs introduced three coupled pan-European intraday auctions (IDAs) in the CWE, Nordic, and Polish bidding zones: IDA 1 at 15:00 CET on D-1, IDA 2 at 22:00 CET on D-1, and IDA 3 at 10:00 CET on D [63]. For the ID auctions, prices are bounded by a price floor of −9999 €/MWh and a price cap of 9999 €/MWh [59].

2.3. Balancing markets (IB)

After the gate closure of the ID market, BRPs must resubmit their balanced position to the TSO. However, deviations can still occur between scheduled and actual generation or consumption. These deviations result from various factors, including plant outages, unexpected demand fluctuations, changes in RES output in real-time, and differences in time resolutions between the settlement periods of the DA, ID, and IB markets [30,64]. In European markets, TSOs manage these real-time frequency deviations through IB markets, employing two key mechanisms: procurement, which involves reserving and activating reserve capacity, and settlement, where TSOs reconcile individual imbalances with BRPs by applying imbalance prices to their positions. IB markets in the EU are cleared by TSOs within their respective control areas, with balancing capacity typically procured before the DA market to ensure sufficient capacity for potential activation in real-time [57].

The TSO calculates the total system imbalance (SI) from the aggregated individual imbalances of BRPs, representing the overall need for flexibility in the system. Following a stepwise procedure, the TSO activates the fastest frequency containment reserves (FCR) first, aiming for full FCR power delivery within 30 s if necessary. For larger deviations, frequency restoration reserves (FRR) come into play, comprising both automatic (aFRR) and manual (mFRR) reserves. aFRR must be capable of full activation within 7.5 min, while mFRR must achieve full activation within 15 min. These reserves can stabilize grid frequency for durations ranging from minutes to hours, depending on the severity of

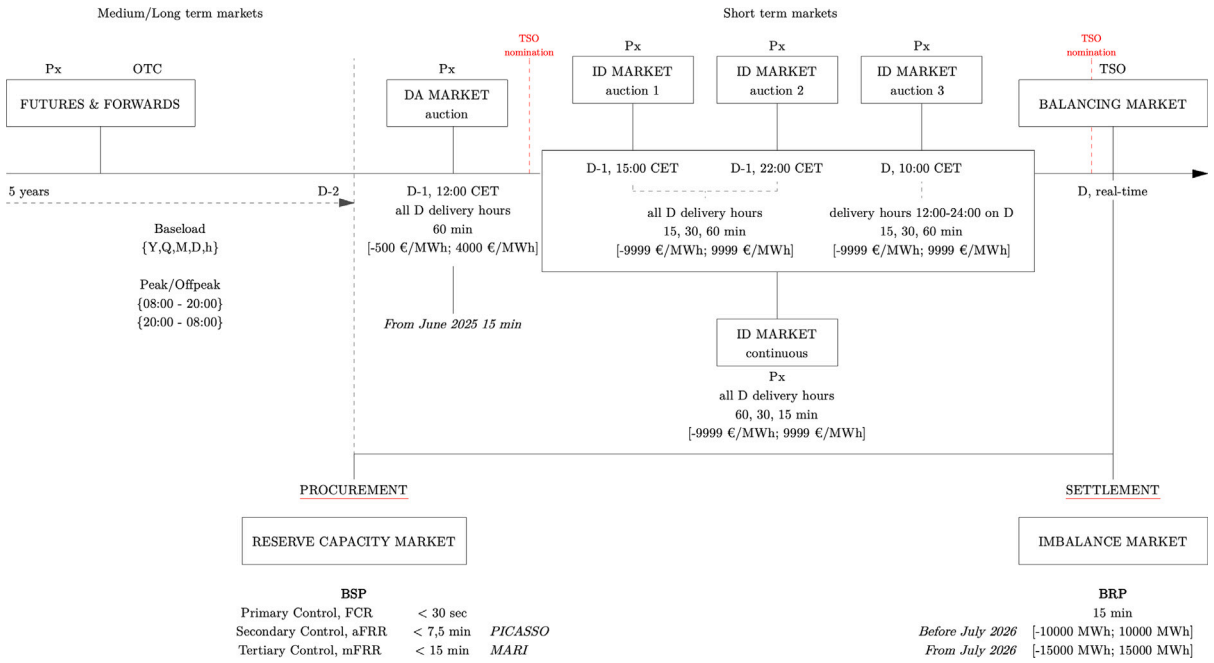


Fig. 3. Schematic overview of whole-sale electricity markets landscape.

the imbalance. Depending on the imbalance's sign, either upward or downward regulation is performed [30]. BRPs also have the capability to address unexpected power variations by activating flexibility within their own portfolio, rather than relying on the TSO to provide flexibility during the settlement of the balancing market. During each settlement period, BRPs can hold a long, short, or balanced position. A long position indicates a positive imbalance, where more power is injected and/or less is withdrawn than planned. Conversely, a short position indicates a negative imbalance, with less power injected and/or more withdrawn than planned. The TSO determines both long and short imbalance prices, with BRPs receiving the long imbalance price for a positive imbalance in real-time and paying the short imbalance price for a negative imbalance [65]. Imbalance prices can be determined using either a dual or single-pricing scheme. In a dual-pricing scheme, the imbalance price for BRPs' positions aligned with the SI is based on reserve capacity activation costs, while those opposing the SI typically follow the DA price. Conversely, a single-pricing scheme applies a uniform imbalance price based on reserve capacity activation costs to all BRPs with imbalanced positions. France employs a dual-pricing scheme, while Belgium, Germany, and the Netherlands adopt a single-price approach [57,64]. The imbalance pricing mechanisms vary across different European countries [66]. High imbalance prices serve as market signals indicating a relative scarcity of inexpensive upward flexibility during negative system imbalances, while negative imbalance prices signal a scarcity of affordable downward flexibility during positive system imbalances [3,57].

Overall, the net regulation volume as a percentage of consumption has been declining in recent years. This reduction is attributed to improved accuracy in RES forecasts, better profiling in the day-ahead and intraday markets, increased liquidity in the intraday market, enhanced market coupling, and more passive balancing by BRPs [3].

3. Flexibility in short-term markets

Currently, flexibility is primarily used within IB markets (e.g., ancillary services) rather than in DA and ID markets. The presence of high amounts of uncertainty in the power system augments the likelihood of activation of expensive fast-response units and the occurrence of load shedding events. This aspect is anticipated to become even more significant in the future, because of increasing penetration and uncertain demand patterns. While DA and ID markets focus primarily on forward contracting and scheduling, IB markets address immediate imbalances between supply and demand. To mitigate this risk, both market operators and the TSO allocate sufficient flexibility in reserve capacity markets [13]. Distributing more flexibility across the DA and ID markets would spread the risk more evenly and reduce the need for intervention by TSOs, leading to more efficient energy utilization. The following sections explore measures aimed at enhancing market design to maximize flexibility utilization by spreading it across multiple market segments and thereby lowering the net regulation volume. The first two aspects, pertaining to time and space, revolve around organizational changes in market design to enhance flexibility, while the third aspect, demand-side response, focuses on how flexibility can mitigate uncertainty through price signals.

3.1. Time

3.1.1. Time granularity of DA-market

The need for more granular and differentiated price signals is growing, with differences in settlement periods across markets. While the DA market operates with an hourly resolution, the ID market offers more frequent options (e.g., hourly, semi-hourly, and quarterly-hourly products), and IB market settles every quarter-hour. The hourly resolution in the DA market fails to fully represent the dynamics of RES and demand because it averages out fluctuations over the hour, thereby eliminating potential spikes and resulting in inefficiencies. Electricity prices should be determined at a very granular temporal level [4]. Implementing a higher granularity (e.g., intervals of 15 min, 5 min, or even second by second) enhances the market's ability to capture the variability of the power system and better represent the flexibility or inflexibility of market participants, reducing the necessity for costly adjustments in the ID market and reserve procurement and activation. Such granularity is particularly beneficial for markets with high penetration of RES [20,21]. A highly granular wholesale market is currently in place in Australia where the market is cleared every 5 min [67]. An alternative approach could be the introduction of a power-based DA market, replacing the current energy-based DA market. This would enable market participants to trade products that better reflect their technical

characteristics and allow the TSO to more accurately account for short-term fluctuations in consumption and production, as well as identify the flexibility available in the market [68,69]. Yet, transitioning to finer temporal resolutions may incur higher computational and data procurement costs. Starting on 11 June 2025, all SDAC markets will transition to 15 min products. This change aims to improve market operation precision, facilitate better integration of renewable energy sources, and enhance overall market efficiency and flexibility. By improving the system's ability to adapt to fluctuations in energy generation and consumption, the new structure will enable more accurate pricing and scheduling, ultimately benefiting market participants [51].

3.1.2. Liquid ID market

The ID market allows market participants to fine-tune their positions from the DA market, often in response to more accurate forecasts. This adjustment can help reduce the requirement for balancing services once again. Continuous trading, which typically operates on a pay-as-bid remuneration scheme, often exhibits relatively low liquidity. This diminished liquidity can discourage participation due to the associated high transaction costs and limited trading opportunities [7]. The continuous trading model fosters a competitive environment often characterized by a "race for speed", creating opportunities for arbitrage [70]. However, to enhance the liquidity of the ID market, transitioning to a series of discrete auctions with a pay-as cleared mechanism, as seen in markets like Spain, Italy, and Portugal, could be beneficial. Unlike continuous bilateral trading, auction mechanisms prioritize the clearance of the most cost-effective bids, reducing the reliance on bids solely due to their availability at specific times. This shift can enhance liquidity indicators by lowering perceived participant risk [7]. Moreover, discrete auctions can mitigate price volatility and establish a reference price, further bolstering liquidity. From June 13, 2024, NEMOs introduced three coupled pan-European intraday auctions (IDAs) in the CWE, Nordic, and Polish bidding zones: IDA 1 at 15:00 CET on D-1, IDA 2 at 22:00 CET on D-1, and IDA 3 at 10:00 CET on D. These multiple IDAs aim to increase liquidity and facilitate the sharing of energy generation resources [63]. Conversely, continuous trading provides greater trading flexibility, allowing transactions to occur immediately rather than waiting for the next market clearing session [3].

3.1.3. Co-optimization of energy and reserves

The impact of multiple reserve procurement mechanisms in Europe is examined in [13]. Reserve procurement and wholesale markets are typically organized as two separate consecutive markets in Europe [31]. This means that capacity allocated to the reserve capacity market cannot be offered in the DA market, leading to suboptimal price formation, as market participants implicitly factor in the opportunity costs of not participating in the DA market when determining reserve capacity prices. This increases the likelihood of inefficient allocation of generation resources [23]. Clearing energy and reserve capacity simultaneously in the same auction, a practice already implemented in the Australian NEM (National Electricity Market) and several North American markets, would lead to a reduction in scheduled reserve capacity. Additionally, it would optimize resource utilization, decrease curtailment levels of RES, mitigate the likelihood of load shedding events, and minimize overall costs [13,26,71,72]. Furthermore, this approach enhances trading opportunities for market participants, fosters more efficient bidding strategies, improves price formation, and provides greater flexibility to market participants [23,73]. However, this approach necessitates reorganizing the roles of market operators, given that in European markets DA and reserve markets are typically cleared by separate entities, power exchanges for the DA market and the TSO for the reserve market.

3.1.4. Smart orders

An important aspect of electricity market clearing involves the bid format used by market participants to provide essential information for clearing the market [13]. The conventional bid format, comprising a quantity [MWh] and price [€/MWh] pair per time interval, might be too restrictive to enable full expression of demand flexibility [42]. Smart orders offer diverse flexibility levels considering multiple time intervals for optimizing trading strategies and adapting to market dynamics. Block and complex bids include techno-economic details like ramp constraints and revenue thresholds, ensuring efficient market clearing [50]. Loop blocks offer the bundling of buy and sell orders, particularly useful for representing storage activities. Flexibility lies in the pairing of these blocks, allowing participants to manage storage efficiently. Storage orders enable market participants to exchange energy across different periods of the market-clearing horizon. These bids can be extended to include arbitrage bids, wherein storage facilities capitalize on price discrepancies over time. Parameters such as storage levels, efficiency loss, and charging and discharging rates are considered in these bids [42]. Curtailable blocks provide options for full execution or rejection based on predefined criteria. Participants can also set minimum acceptance ratios, adding another layer of flexibility to their trading strategies. Linked blocks introduce a dependency on the execution of parent blocks, adjusting this linkage offers flexibility to adapt to changing market dynamics. Exclusive blocks, on the other hand, limit execution to only one block at most, ensuring trading occurs at the most advantageous moment. This setup provides flexibility in timing, allowing participants to prioritize execution based on market conditions or individual strategies. Currently, in some market scenarios, there might be instances where block orders are accepted and rejected simultaneously due to market dynamics or constraints. Implementing adjustable profile blocks could help mitigate this issue by allowing partial execution, effectively eliminating the occurrence of paradoxically accepted and rejected block orders [74]. To encourage higher participation from the demand side, adjustable demand and deferrable demand bids could be introduced [42]. Redefining the requirements and parameters of the bid format to suit the specific needs and capabilities of demand aggregators and storage technologies makes the market more inclusive and adaptable to modern advancements. This adaptation facilitates the seamless integration of these participants, enabling them to leverage their unique strengths such as demand response capabilities and energy storage services. As a result, the market can capitalize on the flexibility and synergies offered by these new players, ultimately enhancing market efficiency [13,42].

3.2. Space

3.2.1. Nodal pricing

Another aspect is spatial granularity, where prices vary at a detailed spatial level — potentially at each connection point in the network — to reflect variations in demand or costs across different locations [4]. The use of short-time interval locational pricing has already spread across US power markets, together with wider areas being managed by Independent System Operators (ISOs) but remains underutilized across Europe [4]. So-called locational marginal prices (LMP), which reflect the marginal value of electricity at each node, are recomputed by the System Operation (SO) every five minutes [4,44]. The spatial granularity depends on the market design and congestions in the systems. European electricity markets currently utilize zonal pricing, which simplifies the network into defined zones with unlimited capacity and uniform prices within each zone. However, this model faces challenges due to the growing disparity between network and generation expansion within bidding zones [35]. A shift towards a more detailed locational pricing approach could be a significant improvement to the existing market design [4,32,33,37]. TSOs resolve intra-zonal congestions through re-dispatch actions following the clearance of DA electricity markets. The adoption of nodal pricing could

yield higher socioeconomic welfare compared to alternative models like uniform and zonal pricing, particularly when considering the cost savings resulting from enhanced dispatch efficiency and reduced electricity prices [4].

3.2.2. Local markets

The rapid advancement and cost reduction of emerging digital technologies have accelerated the widespread deployment of distributed energy resources (DER) and empowered end-users to play an active role in the energy landscape [13]. Local markets offer increased spatial granularity for electricity trading, allowing for more localized transactions and optimization of energy resources. Various setups can be implemented to facilitate this, such as a community-based peer-to-peer model. In this model, an energy collective operates on two levels: at the first level, community members trade their energy internally, while at the subsequent level, the community, acting as a BRP, trades any excess energy on the wholesale market. [75]. Alternatively, the local market operator can manage the local resources within the market to participate in the wholesale markets [76]. This presents an opportunity to establish local flexibility markets, wherein the flexibility offered by local participants (typically coordinated by aggregators) is supplied to BRPs to mitigate imbalances, as well as to DSOs for congestion management and voltage regulation purposes [43,77]. Local markets empower end-users by tailoring market operations to their specific characteristics, preferences, and decisions. This customization allows end-users to actively participate in the energy market, offering flexibility to the overall system. For instance, end-users can adjust their energy consumption patterns based on personal preferences or external factors like price fluctuations, thereby contributing to the efficient management of energy resources at the local level. Once more, market liquidity is crucial in this context. The limited engagement of consumers poses a significant challenge for local markets. Consumers might be reluctant to involve themselves in intricate energy-related decisions, often stemming from a lack of awareness, understanding, or interest in energy matters. This reluctance to engage can lead to reduced participation levels in local energy markets, impacting their overall effectiveness and efficiency [78].

3.2.3. Integrated markets

The aim of integrated markets is to establish a single EU cross-zonal electricity market. An integrated market will enhance the overall efficiency of trading by fostering effective competition, increasing liquidity, and enabling more efficient utilization of generation resources across Europe [4,52,79]. Starting from 2014 the DA market evolved to an integrated SDAC with a common day-ahead power price calculation using the PCR solution where it allocates scarce cross-border transmission capacity in the most efficient way by coupling wholesale electricity markets from different regions [49]. As RES increase, market participants are becoming more interested in trading in the ID markets. This interest is driven by the growing difficulty of maintaining balance after the DA market closes. The ability to balance positions up to one hour before delivery is beneficial for both market participants and power systems. It reduces the need for reserves and associated costs, while providing enough time for system operation processes to ensure security. Since June 12, 2018, the Single Intraday Coupling (SIDC) was launched, making it easier for market participants to manage unexpected changes in consumption and outages. Since 2018, the intraday market has been progressively integrated, enabling participants across Europe to trade electricity continuously on the day it is required [80]. The latest enhancement includes integrating three new intraday auctions in the CWE, Nordic, and Polish bidding zones [63]. Given the full harmonization of DA markets, the coupled and harmonized ID markets, the next logical step would involve the IB markets. The Belgian, Dutch, and German balancing markets currently operate on 15-min settlement periods. Meanwhile, French balancing markets, initially operating on 30-min settlement periods, are also transitioning to 15-min settlement periods [81]. This alignment aims to facilitate the development of an integrated European balancing market, which is currently being tested through Project PICASSO ut faces its problems. This project connects aFRR markets that exhibit significant differences in bidding behaviors related to activation prices [30,82]. Since its launch in 2022, initial participants — including the Czech Republic, Austria, and Germany — have not reported major issues with the new calculation. However, problems emerged when Italy joined, due to differing price methodologies. Italy uses a strictly positive merit order, whereas other countries were able to submit low negative bids [83,84].

Project MARI is the European implementation project for the exchange of balancing energy from manually activated Frequency Restoration Reserves (mFRR) [85]. Launched in 2017 by 19 European TSOs, the platform became operational in 2022 — shortly after the EU Agency for the Cooperation of Energy Regulators (ACER) approval of its implementation framework in January 2020 — with initial participation from German TSOs and ČEPS (i.e., the Czech TSO). Since then, additional TSOs have joined, with further integrations planned through 2025, reflecting the EU's broader commitment to integrated and harmonized balancing markets. To support this integration, the ACER has approved amendments to the pricing methodology for balancing energy [86]. During a transitional period until July 2026, balancing service providers must submit bids within a harmonized price range of –10,000 to +10,000 EUR/MWh. Permanent limits of –15,000 to +15,000 EUR/MWh will apply thereafter. These pricing reforms are designed to promote efficient cross-border balancing, prevent extreme market distortions, and support the effective allocation of transmission capacity across Europe [87]

3.3. Demand-side response and energy storage systems

Demand-side flexibility and energy storage are key enablers in maintaining system balance in electricity markets with increasing shares of RES. While these two forms of flexibility have traditionally been considered separately, their growing interconnection — especially in residential and commercial applications — warrants a more integrated perspective. Demand-side response (DSR) allows consumers to adjust energy consumption in response to market signals or price incentives. Though initially dominated by large industrial users, DSR is expanding into the residential sector due to the wider adoption of solar PV, home batteries, heat pumps, and electric vehicles (EVs) [9,88]. Residential flexibility presents challenges such as small-scale contributions, diverse technologies, and unpredictable occupant behavior [89,90]. Aggregators can address these by pooling smaller loads to create a reliable and dispatchable flexibility portfolio [91], thereby reducing reliance on fossil fuel backup and enhancing system responsiveness [92]. Households can participate through time-of-use tariffs — shifting demand to lower-cost periods — or through incentive-based programs offering payments for load reduction during peak periods. Studies project residential electricity bill savings of 7%–12% in advanced economies and up to 20% in developing markets by 2050 [93]. Several case studies confirm the feasibility of residential flexibility contributing to ancillary services like reserve and balancing [15,17,22,94]. As the share of RES increases, ID trading becomes increasingly relevant for correcting DA forecast errors. Demand-side flexibility can be monetized in the ID market through short-term trading, where higher price volatility closer to delivery enhances profit potential [95]. Market participants can use DSR to adjust consumption during delivery periods (MWh-level flexibility), while reserve markets require capacity-based bids (MW-level) with more rigid activation requirements [28]. Despite these differences, participation in ID trading and reserve markets can provide both system-wide benefits — by reducing balancing service volumes — and individual gains through arbitrage on price spreads [18,28]. Energy storage systems (ESS) complement DSR by enabling energy arbitrage—storing electricity during low-demand, low-price periods and discharging when prices are high or system conditions are tight [96,97]. Storage enhances

system stability, reduces peak generation needs, and contributes to price smoothing. Flexible loads such as power-to-heat (P2H) technologies and thermal storage enable demand response to function as an energy storage proxy, adding further value in both energy and reserve markets [19,98]. Both ESS and DSR are increasingly active in reserve markets, offering fast-responding frequency and voltage regulation [99]. Studies show that reserve market participation often yields higher revenues than energy markets for these resources [16]. To unlock their full potential, regulatory frameworks must evolve. Key enablers include aggregator access, interoperable ICT infrastructure, transparent market signals, and appropriate compensation mechanisms for both energy and flexibility services [19,96].

3.4. Additional remarks

3.4.1. Regulatory sandboxes to test new market designs

Regulatory sandboxes are widely regarded as an important instrument to allow policy and regulation to evolve with the changes in our energy system and to create a level playing field for new technologies and business models that emerge with the energy transition. These sandboxes offer a way to experiment with regulation, where energy actors can propose innovative projects that currently face regulatory barriers. Following the approval of a sandbox project, regulatory derogations are granted, allowing these new technologies or business models to be tested in a real-world environment. In this way, regulatory sandboxes aim to promote innovation and inspire new regulations to support the energy transition. Various studies emphasize the potential of regulatory sandboxes for fostering innovation in electricity market structures. The necessity of regulatory sandboxes to regulate radical innovations in the EU electricity markets is raised in [100,101]. They argue that current EU legislation is insufficient for innovations that do not comply with existing rules and advocate for an EU framework for regulatory sandboxes to test these innovations in a real environment with regulatory safeguards and support. This could lead to changes in legislation [100]. The latter study also investigated experiences with regulatory sandboxes in various European countries and concluded that the outcomes of existing sandbox projects validate the idea of using regulatory sandboxes to promote innovation [100]. Another study describes regulatory sandboxes as participatory experimentation environments for exploring revisions to energy [102]. These sandboxes facilitate a two-way dialogue between an experimenter and a regulator to innovate regulation and enable new socio-technical arrangements [102]. A recent systematic literature review concludes that regulatory sandboxes and pilot projects in the energy sector have emerged as essential instruments to stimulate innovation, facilitate informed regulatory changes, and help achieve decarbonization goals [103]. In summary, the aforementioned studies emphasize that regulatory sandboxes form a valuable instrument to stimulate innovation in electricity market structures by offering flexibility in regulation, facilitating experiments with new technologies and business models, and creating a dialogue between innovators and regulators.

3.4.2. Cross-sectoral integration

Sector coupling — such as the interaction between heating, transportation, and hydrogen sectors — enhances flexibility in short-term electricity markets by balancing supply and demand fluctuations from RES [104,105]. Technologies such as power-to-gas (P2G), power-to-heat (P2H), and transport electrification help integrate intermittent RES, reducing curtailment and price volatility [104,106]. Power-to-Gas (P2G) converts surplus electricity into hydrogen or synthetic methane, which can be stored and later used for electricity generation, heating, or transport [104]. P2G absorbs excess renewable electricity, mitigating curtailment while providing backup power during peak demand [106]. Flexible hydrogen production is expected to play a dominant role in balancing energy systems, particularly for seasonal fluctuations [105]. Power-to-Heat (P2H) uses electric boilers and heat pumps to convert surplus electricity into thermal energy for district and residential heating [104] This helps stabilize the grid by increasing demand during high RES production periods [106]. Thermal storage further enhances flexibility by decoupling heat production from electricity consumption, ensuring efficient energy use [107]. Transport electrification introduces new flexible loads through battery electric vehicles (BEVs), which can be charged to absorb surplus renewable electricity [106,108]. Vehicle-to-grid (V2G) technology further enhances flexibility by enabling EVs to supply electricity back to the grid [106]. Recent studies emphasize cross-sector integration for electricity market flexibility. It has been demonstrated that combining flexibility from transport, heating, and hydrogen optimizes system adequacy and operational costs [106]. BEVs, P2G, and thermal energy storage smooth renewable variability and reduce system costs [108]. Ultimately, sector coupling is essential for integrating energy vectors, improving demand response, and ensuring cost-effective decarbonization [106].

3.4.3. Energy communities

Local energy communities (LECs), also known as citizen energy communities (CECs) or renewable energy communities (RECs), enable citizens to participate in energy generation, consumption, storage, and trading. These communities operate through cooperatives or municipal corporations, aiming to provide affordable energy and integrate new technologies. European legislation acknowledges their role in the energy transition, allowing joint generation, aggregation, and peer-to-peer (P2P) trading [109]. Decentralized energy trading within LECs optimizes local production and consumption, reducing reliance on the central grid and enhancing self-sufficiency [110]. P2P trading allows direct energy exchange among community members, offering participants greater control over their energy use [110]. Various models for P2P trading have been explored, including bilateral trading and centrally coordinated models within the community [111]. LECs also provide flexibility solutions to address uncertainties in renewable generation. High shares of RES, such as solar and wind, introduce operational challenges in distribution networks [43,110]. Energy communities enhance flexibility through DSR, load shifting, and energy storage [43]. Prosumers with flexible assets such as batteries, heat pumps, and EVs can adjust consumption to match renewable generation [43]. Local flexibility markets (LFMs) facilitate the trade of community flexibility with DSOs for congestion management and grid stability [43,110]. By utilizing local flexibility, renewable variability can be managed at the community level rather than impacting the broader grid. Aggregators play a key role by bundling flexibility from smaller participants, improving their market access [109,110]. LECs, through decentralized trading and flexibility services, contribute significantly to renewable energy integration and a resilient energy system [110].

4. Conclusion

Due to the ongoing integration of RES, the variability in the system is increasing. The importance of flexibility in electric energy supply and demand is prominently highlighted in short-term markets. Allocating more flexibility across DA and ID markets would distribute the risk more evenly and diminish the necessity for TSO intervention, thereby enhancing energy utilization efficiency. A comprehensive understanding of the structure of short-term markets, alongside emerging trends, is crucial for assessing the demand for and availability of flexibility in these markets.

Initially, the paper explored the progression of wholesale electricity markets, particularly emphasizing short-term markets. While medium to long-term futures markets facilitate trading over extended periods, short-term markets, encompassing DA, ID, and IB markets, are instrumental in regulating energy supply and demand dynamics. Focusing on the DA market, the paper examined its operational aspects, including available contracts, time resolutions, cross-border trading,

price limits, and trading granularity. The DA market functions through daily auctions, establishing prices for the subsequent day's delivery via hourly trading. It utilizes a single price coupling algorithm to determine market-clearing prices, striving to optimize welfare and transparency. The ID market offers opportunities for adjusting positions closer to delivery and managing forecast uncertainties. This market comes in two primary forms: continuous trading and auction trading, each offering distinct features essential for optimizing short-term market activities. IB markets rectify discrepancies between nominated and actual generation or consumption. Through reserve procurement and settlement mechanisms, this market aims to balance the system promptly. Advancements in market coupling, forecasting accuracy, and increased liquidity have significantly enhanced the efficiency of these markets, particularly in accommodating the growing presence of RES.

Next, the paper discusses several strategies to enhance the flexibility of short-term electricity markets across three dimensions: time, space, and demand response. These strategies aim to address the challenges posed by the increasing penetration of RES and the variability in demand patterns. Firstly, there is a focus on improving the time granularity of market settlements. By moving to shorter intervals, like 15-min periods, markets can better capture the fluctuations in RES generation and demand, leading to more accurate pricing and reduced need for costly adjustments. Another approach involves transitioning to discrete auctions with a pay-as-cleared mechanism in the ID market. This change, already seen in some markets, enhances liquidity and reduces price volatility, improving overall market efficiency. Furthermore, there is a push for co-optimizing energy and reserve markets. By clearing these simultaneously, it is possible to optimize resource utilization, minimize curtailment of RES, and reduce overall costs, leading to more efficient market outcomes. Another focus was on smart orders, which redefine bid formats to include various types of orders that offer flexibility to market participants in different time-intervals. These orders adapt to market dynamics and optimize trading strategies, further enhancing market efficiency and flexibility. Spatial granularity is emphasized to reflect variations in demand and costs across different locations. This involves implementing detailed pricing structures that account for these differences, potentially at each connection point in the network. Local markets can also play a role in enhancing flexibility. By allowing for more localized transactions and optimization of energy resources, end-users can actively participate in the market, contributing to overall system flexibility. Integrating markets will enhance overall trading efficiency by fostering effective competition, increasing liquidity, and enabling more efficient utilization of generation resources, ultimately offering greater flexibility. Finally, flexibility is applied within the sequential short-term markets through demand-response mechanisms based on dynamic contracting. The value from a system's perspective is that they can help reduce the volume of activated balancing services and reduce the need for intervention by TSOs, leading to more efficient energy utilization. These strategies aim to improve the efficiency of short-term electricity markets, better integrate RES, and adapt to evolving demand patterns, thereby ensuring a reliable and sustainable energy supply.

Future research could focus on developing integrated models that capture the interactions between flexibility provision, the newly proposed market designs, and price volatility across all three markets. These models should consider various sources of flexibility and examine how their availability and utilization impact market outcomes. By varying factors such as the penetration of renewable energy, the availability of flexible resources, and market design parameters, different scenarios can analyze the influence on price volatility and market efficiency.

In conclusion, this analysis offers valuable insights for policymakers, market operators, and investors navigating the evolving landscape of increased integration of RES in short-term electricity markets. Going beyond mere recognition of flexibility's importance, our exploration delves into strategies aimed at enhancing the recognition, incentivization, and utilization of flexibility resources. By advocating strategically for flexibility, the paper proposes a proactive approach to address uncertainties, equipping market participants with the necessary tools and incentives to contribute effectively to grid stability.

CRediT authorship contribution statement

Lander Mentens: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Herbert Peremans:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Conceptualization. **Johan Springael:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Philippe Nimmegeers:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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